THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4410 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851$ cm³; cube $= 9261$ cm³; Wasted $= 4410$ cm³
2	**(d)** 2425.5 cm³	Cylinder $= \pi r^2(2r) = 7276.5$ cm³; sphere $= \frac{4}{3}\pi r^3 = 4851$ cm³; Unoccupied $= 2425.5$ cm³
3	**(c)** 15 cm	$H = 3h = 3 \times 5$; $H = 15$ cm
4	**(a)** 14 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 14$ cm
5	**(c)** 4 : 9	$2^2 : 3^2 = 4 : 9$; Ratio $= 4 : 9$
6	**(b)** 8 : 27	$2^3 : 3^3 = 8 : 27$; Ratio $= 8 : 27$
7	**(d)** 30	$\pi(6)^2(10) = n \times \frac{1}{3}\pi(3)^2(4)$; $n = 30$
8	**(a)** 643.5 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(9^3 - 7.5^3)$; Volume $= 643.5$ cm³
9	**(a)** Both A and R are true and R is the correct explanation of A.	$2\pi r^2 + \pi r^2 = 3\pi r^2$; R gives the reason.
10	**(c)** A is true but R is false.	$\frac{1}{3}\pi r^2 h$ against $\pi r^2 h$: A is true; the cone volume is $\frac{1}{3}\pi r^2 h$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(6)^3 = \pi(8)^2 x$ (1); $x = 4.5$ cm (1).	1+1
12	Volume of material = outer hemisphere − inner hemisphere; $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(7.5^3 - 4.5^3)$ (1); Volume $= 693$ cm³ (1).	1+1
13	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(12)^3 = \pi(2)^2 L$ (1); $L = 576$ cm (1).	1+1
14	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(9)^3 = \pi(12)^2 x$ (1); $x = 6.75$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume of cylinder $= \pi \times 9^2 \times 32$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 9$ (1).	1+1+1
16	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(10)^2(5)$ gives $R^3 = 125$ (1); $R = 5$ cm (1).	1+1+1
17	Volume of cylinder $= n\times$ volume of one cone (1); $\pi(6)^2(20) = n \times \frac{1}{3}\pi(3)^2(8)$ (1); $n = 30$ (1).	1+1+1
	OR: Volume of cylinder $= \pi \times 6^2 \times 72$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 9$ (1).	1+1+1
18	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(8)^2(4)$ gives $R^3 = 64$ (1); $R = 4$ cm (1).	1+1+1
	OR: Volume of cylinder $= n\times$ volume of one cone (1); $\pi(12)^2(14) = n \times \frac{1}{3}\pi(6)^2(21)$ (1); $n = 8$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2\pi rh = 2 \times \frac{22}{7} \times 3.5 \times 7 = 154$ mm² (1). (ii) Two hemispheres make a sphere: $4\pi r^2 = 4 \times \frac{22}{7} \times 12.25 = 154$ mm² (1). (iii) $154 + 154 = 308$ mm² (1). (iv) Length $= 7 + 3.5 + 3.5 = 14$ mm (1); volume of cylinder $= \frac{22}{7} \times 12.25 \times 7 = 269.5$ mm³ (1).	1+1+1+2
20	Slant height of the cone $l = \sqrt{63^2 + 84^2} = 105$ cm (2); TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$ (2); TSA $= \frac{22}{7} \times 63 \times 105 + 2 \times \frac{22}{7} \times 63^2 = 45738$ cm² (1).	2+2+1
	OR: The cylinder has radius r and height $2r$ (2); Cylinder $= \pi r^2(2r) = 465696$ cm³; sphere $= \frac{4}{3}\pi r^3 = 310464$ cm³ (2); Unoccupied $= 155232$ cm³ (1).	2+2+1

Mistakes to avoid:

- Counting the joined faces in the total surface area of a combination of solids (e.g. adding the base of a cone resting on a hemisphere).
- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm²/cm³ in the answer.

- Using diameter in place of radius while substituting into a formula.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 24 cm	$H = 3h = 3 \times 8$; $H = 24$ cm
2	**(b)** 20 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 20$ cm
3	**(b)** 9 : 25	$3^2 : 5^2 = 9 : 25$; Ratio $= 9 : 25$
4	**(d)** 27 : 64	$3^3 : 4^3 = 27 : 64$; Ratio $= 27 : 64$
5	**(c)** 6	$\pi(7)^2(12) = n \times \frac{1}{3}\pi(7)^2(6)$; $n = 6$
6	**(d)** 5544 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(15^3 - 9^3)$; Volume $= 5544$ cm^3
7	**(d)** 119070 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{63}{2}\right)^3 = 130977$ cm^3; cube $= 250047$ cm^3; Wasted $= 119070$ cm^3
8	**(c)** 65488.5 cm³	Cylinder $= \pi r^2(2r) = 196465.5$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 130977$ cm^3; Unoccupied $= 65488.5$ cm^3
9	**(d)** A is false but R is true.	Volume is conserved, not surface area (e.g. 8 small spheres have twice the surface area of the big one): A false; R true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$l = \sqrt{36 + 64} = 10$ so A is true; R is also true but has no connection with the slant height.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of material = outer hemisphere − inner hemisphere; $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(14^3 - 3.5^3)$ (1); Volume $= 5659.5$ cm^3 (1).	1+1
12	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(9)^3 = \pi(1)^2 L$ (1); $L = 972$ cm (1).	1+1
13	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(12)^3 = \pi(8)^2 x$ (1); $x = 36$ cm (1).	1+1
14	Volume of material = outer hemisphere − inner hemisphere; $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(13.5^3 - 6^3)$ (1); Volume $= 4702.5$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume of cylinder $= \pi \times 12^2 \times 24$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 12$ (1).	1+1+1
16	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(5)^2(20)$ gives $R^3 = 125$ (1); $R = 5$ cm (1).	1+1+1
17	Volume of cylinder $= n\times$ volume of one cone (1); $\pi(8)^2(9) = n \times \frac{1}{3}\pi(4)^2(6)$ (1); $n = 18$ (1).	1+1+1
	OR: Volume of cylinder $= \pi \times 12^2 \times 16$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 8$ (1).	1+1+1
18	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(10)^2(40)$ gives $R^3 = 1000$ (1); $R = 10$ cm (1).	1+1+1
	OR: Volume of cylinder $= n\times$ volume of one cone (1); $\pi(5)^2(24) = n \times \frac{1}{3}\pi(5)^2(8)$ (1); $n = 9$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $l = \sqrt{7^2 + 24^2} = 25$ m (1). (ii) $\pi rl = \frac{22}{7} \times 7 \times 25 = 550$ m^2 (1). (iii) $2\pi rh = 2 \times \frac{22}{7} \times 7 \times 6 = 264$ m^2 (1). (iv) Total $= 550 + 264 = 814$ m^2 (1); cost $= 814 \times 50 =$ ₹ 40,700 (1).	1+1+1+2
20	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (2); Cylinder $= 62370$ cm^3; two hemispheres $= 38808$ cm^3 (2); Volume $= 101178$ cm^3 (1).	2+2+1
	OR: Remaining volume = cube − hemisphere (2); Cube $= 3176523$ cm^3; hemisphere of radius 73.5 cm $= \frac{2}{3}\pi r^3 = 831946.5$ cm^3 (2); Volume $= 2344576.5$ cm^3 (1).	2+2+1

Mistakes to avoid:

- Counting the joined faces in the total surface area of a combination of solids (e.g. adding the base of a cone resting on a hemisphere).
- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
- Assuming surface area stays constant when a solid is melted and recast; only the volume is conserved.
- Mixing units (cm, m, mm) or writing cm instead of cm^2/cm^3 in the answer.

- Using diameter in place of radius while substituting into a formula.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Surface Areas and Volumes** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 16 : 49	$4^2 : 7^2 = 16 : 49$; Ratio $= 16 : 49$
2	**(a)** 1 : 125	$1^3 : 5^3 = 1 : 125$; Ratio $= 1 : 125$
3	**(b)** 18	$\pi(6)^2(8) = n \times \frac{1}{3}\pi(2)^2(12)$; $n = 18$
4	**(b)** 2062.5 cm³	$\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(10^3 - 2.5^3)$; Volume $= 2062.5$ cm^3
5	**(d)** 282240 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{84}{2}\right)^3 = 310464$ cm^3; cube $= 592704$ cm^3; Wasted $= 282240$ cm^3
6	**(b)** 155232 cm³	Cylinder $= \pi r^2(2r) = 465696$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 310464$ cm^3; Unoccupied $= 155232$ cm^3
7	**(d)** 9 cm	$H = 3h = 3 \times 3$; $H = 9$ cm
8	**(a)** 42 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 42$ cm
9	**(a)** Both A and R are true and R is the correct explanation of A.	The joined faces are inside the solid and are not counted in the total surface area; R explains why only the two curved surfaces are added.
10	**(c)** A is true but R is false.	$\frac{4}{3}\pi(27) = 36\pi$ so A is true; the surface area is $4\pi r^2$, so R is false.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(12)^3 = \pi(1)^2 L$ (1); $L = 2304$ cm (1).	1+1
12	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(15)^3 = \pi(10)^2 x$ (1); $x = 45$ cm (1).	1+1
13	Volume of material = outer hemisphere − inner hemisphere; $\frac{2}{3}\pi(R^3 - r^3) = \frac{2}{3} \times \frac{22}{7}(9.5^3 - 5^3)$ (1); Volume $= 1534.5$ cm^3 (1).	1+1
14	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(6)^3 = \pi(1)^2 L$ (1); $L = 288$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Volume of cylinder $= \pi \times 9^2 \times 48$ (1); Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 108$ (1).	1+1+1
16	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(20)^2(10)$ gives $R^3 = 1000$ (1); $R = 10$ cm (1).	1+1+1
17	Volume of cylinder $= n\times$ volume of one cone (1); $\pi(14)^2(6) = n \times \frac{1}{3}\pi(7)^2(18)$ (1); $n = 4$ (1).	1+1+1
	OR: Volume of cylinder $= \pi \times 6^2 \times 54$ (1); Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 54$ (1).	1+1+1
18	Volume of sphere = volume of cone (1); $\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(2)^2(8)$ gives $R^3 = 8$ (1); $R = 2$ cm (1).	1+1+1
	OR: Volume of cylinder $= n\times$ volume of one cone (1); $\pi(9)^2(20) = n \times \frac{1}{3}\pi(3)^2(15)$ (1); $n = 36$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{4}{3}\pi(6)^3 = 288\pi$ cm^3 (1). (ii) $\pi(4)^2 h = 288\pi \Rightarrow h = 18$ cm (1). (iii) $\frac{288\pi}{\frac{4}{3}\pi(27)} = \frac{288}{36} = 8$ spheres (1). (iv) Total surface area of 8 small spheres $= 8 \times 4\pi(9) = 288\pi$ cm^2 against $4\pi(36) = 144\pi$ cm^2 for the original, i.e. twice as much; volume is conserved but surface area is not (2).	1+1+1+2
20	Slant height of the cone $l = \sqrt{21^2 + 28^2} = 35$ m (2); Canvas = CSA of cylinder + CSA of cone (2); $= 1980 + 2310 = 4290$ m^2 (1).	2+2+1
	OR: The sphere has diameter equal to the edge, so $r = \frac{a}{2}$ (2); Sphere $= \frac{4}{3}\pi\left(\frac{147}{2}\right)^3 = 1663893$ cm^3; cube $= 3176523$ cm^3 (2); Wasted $= 1512630$ cm^3 (1).	2+2+1

Mistakes to avoid:

- Counting the joined faces in the total surface area of a combination of solids (e.g. adding the base of a cone resting on a hemisphere).
- Using the vertical height instead of the *slant* height in the CSA and TSA of a cone.
- Forgetting the $\frac{1}{3}$ in the volume of a cone or using $2\pi r^2$ for the surface area of a full sphere.
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